Problem 11.1

Find the effective interest rate over a three-year period which is equivalent to an effective rate of discount of 8% the first year, 7% the second year, and 6% the third year.

Solution

The relationship between an effective discount rate d and the corresponding effective interest rate i is

$$d = \frac{i}{1+i}, \qquad i = \frac{d}{1-d}.$$

Therefore, the effective interest rates for the three years are

$$\begin{aligned}
i_1 &= \frac{0.08}{1-0.08} = \frac{0.08}{0.92} \approx 0.0869565, \\
i_2 &= \frac{0.07}{1-0.07} = \frac{0.07}{0.93} \approx 0.0752688, \\
i_3 &= \frac{0.06}{1-0.06} = \frac{0.06}{0.94} \approx 0.0638298.
\end{aligned}$$

Let i be the equivalent annual effective interest rate. Equating the accumulation over three years gives

$$(1+i)^3 = (1+i_1)(1+i_2)(1+i_3).$$

Thus,

$$(1+i)^3 = \frac{1}{0.92}\frac{1}{0.93}\frac{1}{0.94} \approx 1.2433728.$$

Solving for i,

$$i = (1.2433728)^{1/3} - 1 \approx 0.0753103.$$

Hence,

$$\boxed{i \approx 7.53\%}.$$